



ESTIMATION AND INTERPRETATION OF SERUM CALCIUM

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INTRODUCTION & CLINICAL SIGNIFICANCE

•Role of Calcium:

- One of the most abundant and important minerals in the human body.
- Approximately 99% of body calcium is found in bones.

•Interpretation:

- A decrease in albumin levels can cause a decrease in serum calcium.
- Caution:** A clinical diagnosis should not be made based on a single test result; it must be integrated with other clinical and laboratory data.

•Normal Reference Ranges :

Total Serum Calcium: 8.5–10.5 mg/dL (or 2.1–2.6 mmol/L)

CAUSES OF ABNORMAL CALCIUM LEVELS

- **Hypercalcemia (High Calcium):**

- Cancers
- Large intake of vitamin D
- Enhanced renal retention
- Osteoporosis
- Sarcoidosis
- Thyrotoxicosis
- Hyperparathyroidism

- **Hypocalcemia (Low Calcium):**

- Hypoparathyroidism
- Pseudohypoparathyroidism
- Vitamin D deficiency
- Malnutrition
- Intestinal malabsorption

ESTIMATION OF TOTAL SERUM CALCIUM

1. Objective

To determine the concentration of total calcium in a serum sample using the **o-Cresolphthalein Complexone (OCPC)** colorimetric method.

2. Materials Required

Reagents: Working Reagent (equal parts Buffer and Color Reagent), Calcium Standard (8 mg/dL).

Specimen: Fresh serum or heparinized plasma.

Equipment: Spectrophotometer (set to 578 nm), 1 cm cuvettes, micropipettes (20 μ L and 1000 μ L), and timer.

3. PROCEDURE

A. Reagent Preparation

Mix equal volumes of **Buffer (R1)** and **Color Reagent (R2)** to create the **Working Reagent**.

Allow it to sit at room temperature for **10 minutes** before starting the assay.

B. Pipetting Scheme

Label three clean test tubes or cuvettes as **Blank**, **Standard**, and **Sample**.

CONTINUED....

Step	Blank	Standard	Sample
Working Reagent	1000 \mu L	1000 \mu L	1000 \mu L
Standard	—	20 \mu L	—
Sample (Serum)	—	—	20 \mu L

CONTINUED....

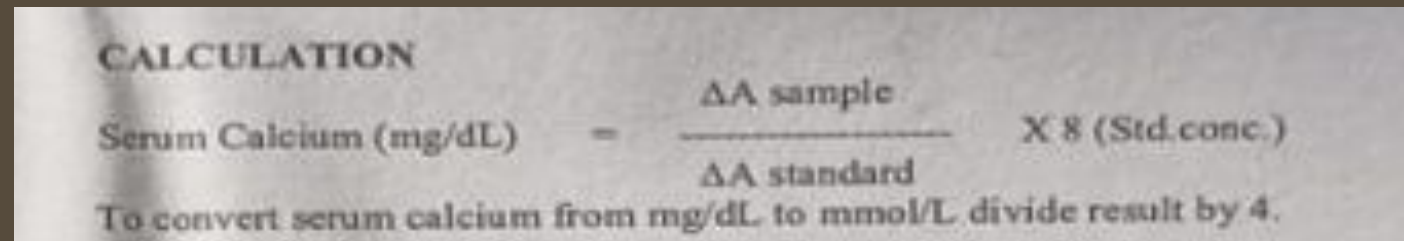
C. Measurement

1. Mix the contents of each tube gently.
2. Incubate at room temperature (20-25 degree C) for **5 minutes**.
3. Set the spectrophotometer to **578 nm** and zero it using the **Blank**.
4. Read and record the absorbance (Delta A) of the **Standard** and the **Sample** within 20 minutes.

4. CALCULATION

- Formula:

Use the following formula to find the concentration:



CALCULATION

$$\text{Serum Calcium (mg/dL)} = \frac{\Delta A \text{ sample}}{\Delta A \text{ standard}} \times 8 \text{ (Std. conc.)}$$

To convert serum calcium from mg/dL to mmol/L divide result by 4.

- Unit Conversion:

To convert serum calcium from mg/dL to mmol/L, divide the result by 4.

5. RESULTS & INTERPRETATION

- **Observed Value:** _____ mg/dL
- **Normal Range (Harper's Biochemistry):** 8.5-10.5 mg/dL
- **Conclusion:** Determine if the sample is **Normocalcemic**, **Hypercalcemic** (high), or **Hypocalcemic** (low).

6. PRECAUTIONS

- **Avoid Contamination**: Calcium is everywhere. Use disposable plastic tubes or acid-washed glassware to prevent false high readings.
- **Lipemia**: If the sample is cloudy (lipemic), a "Sample Blank" (sample + water) must be run to subtract the background turbidity.
- **Safety**: Reagents contain sodium azide; avoid contact with skin and do not swallow.

Here is the link of the calcium test video.

<https://youtu.be/49vRxxC6jl8?si=GJY-A0donRWQxjL6>

thank you!

